Supplementary Material

From Flow Records to Original Connections: Extraction-Aware Auditing of Frozen IoT Intrusion Detectors

Tables S1-S8 provide supporting results for the extraction-aware audit. Model banks, reference cohorts, operating points, alarm policies, and observation horizons remain distinct across tracks. Repeated model, seed, ordering, and window evaluations are dependent appearances, not independent original connections or deployments. NA denotes an undefined or non-evaluable quantity, not zero. Rates are fractions unless a percentage is explicitly indicated; attribution intervals are not sampling confidence intervals.

S1. Capture-48 packet-membership diagnosis

The 10-s condition uses the fixed 512-malicious-unit cohort. MG denotes the whole-membership gate; unresolved units remain in the detection denominator.

Table S1. Membership and associated-record checks at 10 s

Property	Validated units	Unresolved units
Malicious reference units	263	249
Mean expected packet occurrences	2.000	5.904
Mean mapped packet occurrences	2.000	5.904
Unresolved packet-occurrence assignments	0	0
Mean associated records	2.000	2.976
Mean surplus packet count	0	2.028
Units sharing a UID with another selected unit	0	2

Of 269 units with two expected packet occurrences, 263 validate and six remain unresolved. All 243 units with at least six expected occurrences remain unresolved. Among the 249 unresolved units, surplus counts are two for 246 units, four for two units, and five for one unit. Associated-record counts are two for six unresolved units and three for 243 unresolved units. No expected selected-unit occurrences are missing from the retained mapping; this is not a global packet-loss claim. The physical owners of all surplus packets are not established.

S2. Zeek timer-count controls

The sweep comprises 32 replays: two captures, four delays, and four one-factor settings. Each input prefix ends at 32 MiB or 400,000 packets, whichever occurs first.

Table S2a. Timer configurations

Setting	SYN timeout (s)	Attempt delay (s)	Inactivity timeout (s)
DEFAULT	5	5	300
SYN_15S	15	5	300
ATTEMPT_15S	5	15	300
INACTIVITY_15S	5	5	15

Table S2b. Total exported connection records

Capture	Delay (s)	DEFAULT	SYN_15S	ATTEMPT_15S	INACTIVITY_15S
48	0	86,060	86,060	43,661	86,076
48	1	86,062	86,062	43,661	86,078
48	3	86,435	86,435	43,662	86,451
48	10	129,278	129,278	85,719	129,294
49	0	125,847	125,847	125,832	125,858
49	1	125,849	125,849	125,832	125,860
49	3	125,861	125,861	125,833	125,872
49	10	251,514	251,514	125,840	251,525

These counts describe exported records from the bounded input prefixes, not independently identified malicious reference connections. No detector is scored in this sweep. Changed counts do not establish detection recovery for the separate 512-unit cohort.

S3. Paired extractor-count comparison

Zeek 8.0.10 and NFStream 6.6.0 process the same eight capture/delay inputs. Eight Zeek extractions and sixteen NFStream extractions produce 24 result rows. Paired inputs have identical packet hashes.

Table S3. Total exported records by extractor and setting

Capture	Delay (s)	Zeek default	NFStream idle 15 s	NFStream idle 300 s
48	0	86,060	43,361	43,295
48	1	86,062	43,361	43,295
48	3	86,435	43,361	43,295
48	10	129,278	43,361	43,295
49	0	125,847	125,835	125,733
49	1	125,849	125,835	125,733
49	3	125,861	125,835	125,733
49	10	251,514	125,835	125,733

NFStream uses an active timeout of 1,800 s. Its TCP-group record totals remain unchanged across delays: capture 48 has 43,259 (idle 15 s) and 43,219 (idle 300 s); capture 49 has 125,802 and 125,700, respectively. The inputs contain 43,219 TCP groups for capture 48 and 125,669 for capture 49. Expiration semantics differ across extractors, even at equal nominal idle settings. Stable totals do not establish identical features, packet attribution, attack recall, or detection superiority.

S4. Matched MD3 warning-stage comparison

CF denotes continuous first-request MD3; WR denotes the window-reset diagnostic. Condition is the native warning; Condition Only requires a condition warning absent from its clean baseline. Each ordering has sixteen rows; the three subtables contain all 48 summary rows.

Table S4a. Historical hash: both variants and warning rules

Variant	Delay (s)	Warning rule	Eligible	Paired	Coverage	TP	FN	FP	TN	TPR	FPR
CF	0	Condition	24	24	1	0	0	3	21	NA	0.125
CF	0	Condition Only	24	24	1	0	0	0	24	NA	0
CF	1	Condition	24	24	1	0	0	3	21	NA	0.125
CF	1	Condition Only	24	24	1	0	0	0	24	NA	0
CF	3	Condition	24	24	1	0	0	0	24	NA	0
CF	3	Condition Only	24	24	1	0	0	0	24	NA	0
CF	10	Condition	24	24	1	3	21	0	0	0.125	NA
CF	10	Condition Only	24	24	1	0	24	0	0	0	NA
WR	0	Condition	24	24	1	0	0	0	24	NA	0
WR	0	Condition Only	24	24	1	0	0	0	24	NA	0
WR	1	Condition	24	24	1	0	0	0	24	NA	0
WR	1	Condition Only	24	24	1	0	0	0	24	NA	0
WR	3	Condition	24	24	1	0	0	0	24	NA	0
WR	3	Condition Only	24	24	1	0	0	0	24	NA	0
WR	10	Condition	24	24	1	0	24	0	0	0	NA
WR	10	Condition Only	24	24	1	0	24	0	0	0	NA

Coverage and rates are fractions. TPR = TP/(TP + FN); FPR = FP/(FP + TN). NA denotes a zero denominator. Delay 10 s supplies harmful appearances; delays 0/1/3 s supply no-damage controls, not benign-only traffic. The MD3-named hash is an ordering, not an estimator. CF stops at the first request for oracle labels; subsequent unavailable windows are not counted as silent negatives. WR restarts from the same source reference per window and is not the continuous algorithm. Reused seeds and orderings are dependent appearances.

S4. Matched MD3 warning-stage comparison

CF denotes continuous first-request MD3; WR denotes the window-reset diagnostic. Condition is the native warning; Condition Only requires a condition warning absent from its clean baseline. Each ordering has sixteen rows; the three subtables contain all 48 summary rows.

Table S4b. MD3-named hash: both variants and warning rules

Variant	Delay (s)	Warning rule	Eligible	Paired	Coverage	TP	FN	FP	TN	TPR	FPR
CF	0	Condition	24	24	1	0	0	3	21	NA	0.125
CF	0	Condition Only	24	24	1	0	0	0	24	NA	0
CF	1	Condition	24	24	1	0	0	3	21	NA	0.125
CF	1	Condition Only	24	24	1	0	0	0	24	NA	0
CF	3	Condition	24	24	1	0	0	0	24	NA	0
CF	3	Condition Only	24	24	1	0	0	0	24	NA	0
CF	10	Condition	24	24	1	3	21	0	0	0.125	NA
CF	10	Condition Only	24	24	1	0	24	0	0	0	NA
WR	0	Condition	24	24	1	0	0	0	24	NA	0
WR	0	Condition Only	24	24	1	0	0	0	24	NA	0
WR	1	Condition	24	24	1	0	0	0	24	NA	0
WR	1	Condition Only	24	24	1	0	0	0	24	NA	0
WR	3	Condition	24	24	1	0	0	0	24	NA	0
WR	3	Condition Only	24	24	1	0	0	0	24	NA	0
WR	10	Condition	24	24	1	0	24	0	0	0	NA
WR	10	Condition Only	24	24	1	0	24	0	0	0	NA

Coverage and rates are fractions. TPR = TP/(TP + FN); FPR = FP/(FP + TN). NA denotes a zero denominator. Delay 10 s supplies harmful appearances; delays 0/1/3 s supply no-damage controls, not benign-only traffic. The MD3-named hash is an ordering, not an estimator. CF stops at the first request for oracle labels; subsequent unavailable windows are not counted as silent negatives. WR restarts from the same source reference per window and is not the continuous algorithm. Reused seeds and orderings are dependent appearances.

S4. Matched MD3 warning-stage comparison

CF denotes continuous first-request MD3; WR denotes the window-reset diagnostic. Condition is the native warning; Condition Only requires a condition warning absent from its clean baseline. Each ordering has sixteen rows; the three subtables contain all 48 summary rows.

Table S4c. Saved order: both variants and warning rules

Variant	Delay (s)	Warning rule	Eligible	Paired	Coverage	TP	FN	FP	TN	TPR	FPR
CF	0	Condition	12	12	1	0	0	3	9	NA	0.25
CF	0	Condition Only	12	12	1	0	0	0	12	NA	0
CF	1	Condition	12	12	1	0	0	3	9	NA	0.25
CF	1	Condition Only	12	12	1	0	0	0	12	NA	0
CF	3	Condition	12	12	1	0	0	0	12	NA	0
CF	3	Condition Only	12	12	1	0	0	0	12	NA	0
CF	10	Condition	12	12	1	3	9	0	0	0.25	NA
CF	10	Condition Only	12	12	1	0	12	0	0	0	NA
WR	0	Condition	12	12	1	0	0	0	12	NA	0
WR	0	Condition Only	12	12	1	0	0	0	12	NA	0
WR	1	Condition	12	12	1	0	0	0	12	NA	0
WR	1	Condition Only	12	12	1	0	0	0	12	NA	0
WR	3	Condition	12	12	1	0	0	0	12	NA	0
WR	3	Condition Only	12	12	1	0	0	0	12	NA	0
WR	10	Condition	12	12	1	0	12	0	0	0	NA
WR	10	Condition Only	12	12	1	0	12	0	0	0	NA

Coverage and rates are fractions. TPR = TP/(TP + FN); FPR = FP/(FP + TN). NA denotes a zero denominator. Delay 10 s supplies harmful appearances; delays 0/1/3 s supply no-damage controls, not benign-only traffic. The MD3-named hash is an ordering, not an estimator. CF stops at the first request for oracle labels; subsequent unavailable windows are not counted as silent negatives. WR restarts from the same source reference per window and is not the continuous algorithm. Reused seeds and orderings are dependent appearances.

S5. Legacy warnings and reporting coverage

The harmful populations at 10 s comprise 24 historical-hash, 24 MD3-named-hash, and 12 saved-order seed-window appearances. AC denotes average confidence, DoC difference of confidences, ATC average thresholded confidence, and CBPE confidence-based performance estimation.

Table S5. Warning outcomes by ordering

Estimator	Historical hash: TP, FN	MD3-named hash: TP, FN	Saved order: TP, FN	Paired coverage	No-damage warnings / appearances
AC	4, 20	2, 22	8, 4	1	0/72; 0/72; 0/36
DoC	4, 20	2, 22	8, 4	1	0/72; 0/72; 0/36
ATC	8, 16	8, 16	4, 8	1	0/72; 0/72; 0/36
CBPE (source-Platt)	0, 24	0, 24	0, 12	1	0/72; 0/72; 0/36
Competence-only	20, 4	19, 5	12, 0	1	0/72; 0/72; 0/36
Semantic agreement	20, 4	19, 5	12, 0	1	0/72; 0/72; 0/36
Drift-only	24, 0	24, 0	12, 0	1	0/72; 0/72; 0/36
Combined hard gate	NA	NA	NA	0	NA

TP and FN are count pairs, not ratios. No-damage warning fractions pool the 0/1/3-s conditions and are listed in historical-hash, MD3-named-hash, saved-order sequence. Their denominators are 72, 72, and 36. These are dependent appearances, not independent deployments. Seven reporting configurations have full clean, condition, and paired coverage, including CBPE despite its zero warnings. The hard gate has zero clean coverage, approximately 0.3333 mean condition coverage, zero paired coverage, and 96 clean-reject/condition-accept appearances; its sensitivity and false-warning fraction are undefined.

S6. Capture-35 temporal results

The protocol uses four equal-duration periods, six windows per later period, and nominal levels 0.01 and 0.05. Period-1 published-benign records supply calibration. The protocol was frozen before scoring after preliminary label-count inspection; the reproduced run is not a fresh unexposed confirmation.

Table S6a. Later-window check outcomes and attack-window recall

Model	Nominal level	Eligible windows	With-slack pass	Strict pass	Legacy statistic <= observed TP	Minimum later-period AUROC	Median attack-positive-window recall
HGB	0.01	18	18/18	18/18	18/18	0.556459	0.114741
HGB	0.05	18	18/18	18/18	18/18	0.556459	0.114741
LR	0.01	18	18/18	15/18	15/18	0.674232	0.713449
LR	0.05	18	17/18	14/18	15/18	0.674232	0.714080

Table S6b. Period composition and threshold-independent AUROC

Period	Records	Benign	Malicious	HGB AUROC	LR AUROC
1	2,024,537	2,024,442	95	0.376848	0.444494
2	2,078,796	2,078,796	0	NA	NA
3	2,985,051	2,076,504	908,547	0.556459	0.674232
4	3,359,403	2,082,647	1,276,756	0.584294	0.763435

All rates and recall values are fractions. Strict checks compare empirical benign FPR with the calibration envelope; with-slack counts use the saved additional evaluation tolerance. These empirical checks do not establish the sampling assumptions. Window-median recall and period-level AUROC are different aggregates. Period 2 has no malicious annotations, so recall and AUROC are undefined. The legacy statistic max(0, alerts - N times bound) is a diagnostic, not a certified realized attack count. It exceeds observed true positives in 3/18 LR windows at each nominal level. At nominal level 0.05, zero-based window 12 has 346,530 benign records, zero attacks, 18,997 false alarms, and an asserted lower count of approximately 1,339. False-alarm check compliance does not imply high attack recall.

S7. Raw-only pilot

The pilot uses a separate frozen model bank and reference cohorts. Outcomes follow its native pure-witness and deadline rules; these rows are not interchangeable with the historical or fresh 512-unit tracks.

Table S7. Malicious-unit detection and clean-minus-paced change

Capture	Model	Malicious units	Clean detection	10-s detection	Clean minus 10 s
34	HGB	1,492	0.729223	[0.616622, 0.977882]	[-0.248660, 0.112601]
34	LR	1,492	0.739946	[0.638740, 1]	[-0.260054, 0.101206]
49	HGB	2,001	1	0.331334	[0.668666, 0.668666]
49	LR	2,001	1	1	[0, 0]
7	HGB	2,031	0	0	[0, 0]
7	LR	2,031	0.999015	1	[-0.000985, -0.000985]

Values are fractions. Intervals represent attribution uncertainty, not sampling confidence. Capture 34 contains 2,048 panel units: 1,492 malicious, 521 benign, and 35 unlabelled. At 10 s, 539 malicious and four benign outcomes are unresolved. The change intervals span zero and establish neither improvement nor deterioration. Endpoints retain the reported precision; rounding displayed rates before subtraction can change the last decimal. The very small benign pilot cohorts in captures 49 and 7 do not support specificity claims.

S8. NF-ToN-IoT-v2 family-transfer results

This is a same-parent tabular study, not the paired Zeek/NFStream packet-extraction experiment. Each held-out attack-family panel contains 2,560 records. The semantic views group protocol/service, transport-state, size/volume, and timing/rate features; views and attack-family panels are distinct.

Table S8a. Clean held-out-family classifier outcomes and acceptance coverage

Held-out family	Records	Accuracy (%)	AUROC	Normal FPR (%)	Attack recall (%)	Accepted coverage (%)
Injection	2,560	72.54	0.8296	10.63	55.70	0
XSS	2,560	71.80	0.8463	9.84	53.44	40
Ransomware	2,560	48.01	0.5295	10.00	6.02	0
Backdoor	2,560	44.45	0.1793	11.80	0.70	0
MITM	2,560	66.13	0.5198	8.05	40.31	0

Table S8b. Primary-classifier attack recall under report corruption (%)

Held-out family	Clean	epsilon = 0.10	epsilon = 0.25	epsilon = 0.50
Injection	55.70	51.17	38.83	34.14
XSS	53.44	50.55	45.47	32.34
Ransomware	6.02	5.16	5.00	5.00
Backdoor	0.70	0.70	0.70	0.63
MITM	40.31	39.14	36.56	6.80

Coverage is the reported acceptance diagnostic, not primary-classifier recall or validated packet membership. Accepted-error fields at zero coverage are undefined, not zero error. Epsilon indexes the saved report-corruption conditions; it is not a pacing delay or an established packet-valid perturbation budget. No new model fitting or threshold changes were made during that evaluation. Family panels share one parent representation and are not established independent sampling units, corpora, or deployments.

S8. NF-ToN-IoT-v2 source-view and boundary results

The source-test table characterizes five saved models. These source outcomes are not substituted for held-out-family results.

Table S8c. Source-test metrics

View/model	Records	Accuracy (%)	Normal FPR (%)	Attack recall (%)
Primary	4,000	63.40	5.50	32.30
Protocol/service	4,000	50.18	2.60	2.95
Transport state	4,000	67.85	3.90	39.60
Size/volume	4,000	54.28	3.20	11.75
Timing/rate	4,000	58.45	5.05	21.95

Table S8d. Semantic-view boundary summaries

Quantity	Reported value
Mean natural semantic-view accuracy	0.5620
Declared view-accuracy threshold	0.8
Equal-family mean differential-error gain: ANCH versus AC	-0.02595
Families with positive descriptive gain	Injection only

ANCH is the saved development-monitor identifier; AC denotes average confidence. The reported equal-family gain is negative, and no positive independent clustered gain was established. The summary gain is not a primary-classifier accuracy or recall improvement. Packet validity and NetFlow semantic validity of report-corruption conditions were not established. The outcome tables alone do not establish an implementable acceptance gate, a perturbation construction, or a deployable detector-selection policy.